

BOR, Russian Federation

WMO: 238840

Lat: **61.5806N**Lon: **90.0331E**Elev: **189**StdP: **14.60**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.9	-43.4	-53.5	0.2	-49.8	-47.3	0.3	-43.3	10.7	7.8	9.7	7.9	0.3	70	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	85.0	66.4	81.3	65.2	77.2	63.6	69.1	79.9	67.2	76.9	65.2	74.2	5.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	95.4	72.8	63.7	88.8	70.8	61.6	82.6	69.2	33.5	80.4	31.9	77.1	30.3	74.4	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.0	9.6	8.5	DB	-52.1	90.5	6.8	2.6	-57.1	92.4	-61.1	93.9	-64.9	95.4	-69.8	97.3
(5)				WB	-51.9	72.1	6.8	2.5	-56.8	73.9	-60.8	75.3	-64.6	76.7	-69.5	78.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	-11.6	-3.5	14.6	28.9	42.3	59.1	64.4	57.4	44.7	28.9	4.5	-6.9
	DBStd	29.06	19.03	15.52	13.31	10.60	9.57	9.04	6.58	7.01	7.91	10.49	17.82	18.98
	HDD50	9438	1909	1497	1096	632	281	25	1	15	199	656	1364	1763
	HDD65	13976	2374	1917	1561	1082	705	216	93	255	610	1121	1814	2228
	CDD50	1070	0	0	0	0	42	299	446	243	39	1	0	0
	CDD65	131	0	0	0	0	2	39	73	17	0	0	0	0
	CDH74	1546	0	0	0	0	37	573	765	167	4	0	0	0
	CDH80	439	0	0	0	0	9	171	222	37	0	0	0	0
Wind	WSAvg	3.5	2.6	3.0	3.8	4.4	4.3	3.7	3.3	3.2	3.5	3.8	3.3	2.8
Precipitation	PrecAvg	23.2	1.5	1.1	1.1	1.3	1.9	2.2	2.6	2.9	2.6	2.4	2.1	1.7
	PrecMax	29.9	2.9	2.2	3.0	2.9	4.2	4.6	6.5	5.3	4.9	4.5	3.7	3.0
	PrecMin	15.3	0.1	0.2	0.1	0.3	0.7	0.8	0.3	0.5	0.4	0.6	0.8	0.6
	PrecStd	3.5	0.6	0.4	0.6	0.5	0.8	1.1	1.3	1.2	0.9	0.7	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.5	31.0	46.7	60.7	78.5	89.3	90.2	84.6	71.9	54.3	36.4	31.0
		MCWB	28.6	29.8	37.5	46.5	60.4	66.7	69.5	67.5	59.7	47.2	35.0	30.1
	2%	DB	23.5	26.7	39.5	54.1	71.0	85.0	85.5	78.5	66.1	49.4	33.9	27.1
		MCWB	22.9	25.1	33.1	42.9	55.9	65.5	67.4	65.3	56.5	45.3	33.4	26.2
	5%	DB	18.1	21.9	35.2	48.7	65.2	80.9	82.0	73.9	60.8	45.6	32.0	24.1
		MCWB	17.2	20.6	31.0	40.3	52.7	63.8	66.1	63.0	54.0	42.2	31.3	23.3
	10%	DB	13.4	17.6	32.5	44.0	59.7	76.3	78.2	69.5	56.7	41.9	29.2	19.7
		MCWB	12.6	16.5	29.4	37.7	49.6	62.0	64.9	61.0	52.2	39.4	28.3	18.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.9	30.1	38.0	47.9	62.2	69.9	72.7	71.0	61.3	49.8	35.1	30.4
		MCDB	29.6	31.0	44.3	58.0	77.0	82.6	84.4	80.6	69.0	53.0	35.8	30.9
	2%	WB	22.9	25.2	34.1	44.2	57.5	67.5	70.0	67.0	57.9	45.6	33.5	26.3
		MCDB	23.6	26.5	38.0	52.2	68.9	79.9	81.0	75.3	63.8	48.6	33.9	26.9
	5%	WB	17.3	20.6	31.9	41.1	53.9	65.5	68.2	64.3	55.2	42.8	31.4	23.4
		MCDB	18.0	21.9	35.0	48.0	63.4	76.6	77.8	71.5	59.8	45.3	31.9	24.2
	10%	WB	12.7	16.4	29.3	38.1	50.2	63.5	66.4	62.0	52.5	39.4	28.3	18.9
		MCDB	13.4	17.5	32.5	43.2	58.8	74.1	75.2	68.1	56.4	41.6	29.2	19.7
Mean Daily Temperature Range	5% DB	MDBR	13.8	17.4	20.1	18.5	18.5	20.6	19.1	16.1	12.9	9.5	11.8	13.1
		MCDBR	12.8	16.2	19.9	24.2	28.7	29.7	26.8	23.7	20.8	13.3	9.8	13.1
		MCWBR	12.6	15.4	15.1	15.2	16.2	13.1	11.1	11.5	11.9	9.7	9.9	12.8
	5% WB	MCDBR	13.0	16.1	17.5	22.0	26.4	24.1	22.0	20.1	17.6	12.6	10.1	13.3
		MCWBR	12.8	15.5	13.9	14.6	15.3	12.1	10.5	10.9	11.0	9.7	10.1	13.0
Clear-Sky Solar Irradiance	taub	0.266	0.304	0.340	0.365	0.382	0.391	0.451	0.411	0.357	0.320	0.258	0.240	
	taud	1.914	1.971	1.975	2.040	2.243	2.371	2.173	2.272	2.390	2.235	1.970	1.870	
	Ebn at Noon	120	196	235	257	265	265	242	243	237	197	123	73	
	Edh at Noon	14	26	37	43	38	35	41	34	25	20	13	9	
All-Sky Solar Radiation	RadAvg	108	383	805	1282	1549	1755	1603	1118	661	279	124	51	
	RadStd	15	32	95	152	170	182	142	107	88	48	24	9	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)